TEXORA · LECTURE SHEET

Frictional Properties

The grip between surfaces: Amontons' laws and where fibres break them, the capstan equation that multiplies tension around every guide and roller, stick–slip, fibre-to-fibre cohesion that spinning depends on — and wool's directional friction, the engine of felting.

TE-3209 · TEXTILE PHYSICS · TOPIC 5 · LEVEL 3 TERM 2

1. Friction laws — and fibre exceptions

Amontons' laws state that friction force F is proportional to normal load N (F = μN, defining the coefficient of friction μ) and independent of apparent contact area. They serve metals tolerably; fibres obey them only roughly. Fibre friction is better described by F = aN^n with n often 0.7–1.0, because the real contact area grows as compliant fibres flatten into each other — friction depends on the materials' softness and surface films as much as on load. Fibre friction is also **velocity- and moisture-sensitive**: a water film lubricates some pairs and (by meniscus adhesion) sticks others; spin finishes exist precisely to park μ in a narrow working window — low enough to run, high enough to cohere.

2. The capstan equation — tension multiplication

A yarn or belt wrapping an angle θ (radians) around a fixed surface trades tension across it by the capstan (Euler) relation:

$$T_2 = T_1 \, e^{\wedge}(\mu\theta)$$

where T_2 is the tension on the tight side when motion impends toward T_2. Every guide eye, roller and tension disc in a processing line is a capstan transaction — the fundamento of yarn tension control from winding to weaving.

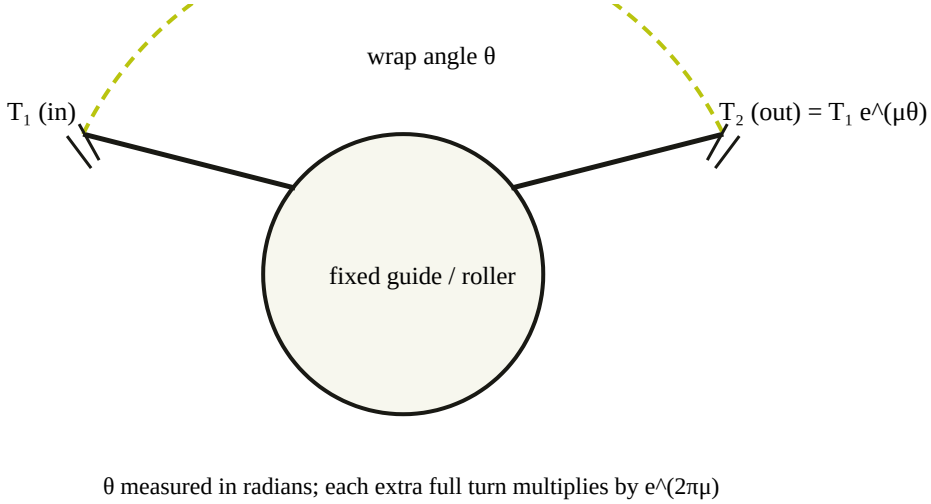


Figure 1 — The capstan arrangement: yarn wrapping a fixed surface through angle θ. Tension grows exponentially toward the direction of impending motion.

WORKED EXAMPLE — GUIDE FRICTION IN A YARN PATH

Yarn with μ = 0.25 against a ceramic guide wraps 180° (θ = π): $T_2/T_1 = e^{0.25\pi} = e^{0.785} =$ **2.19**. An input of 10 cN leaves at 21.9 cN. Add a second guide making the total wrap 270° at μ = 0.3: ratio = e^(0.3 × 4.712) = e^1.414 = **4.11** — the path, not the machine setting, wrote that tension. Conversely the same equation in reverse is the warper's friend: a few turns on a capstan let a small holding force restrain a large load (the sailor's bollard principle).

3. Stick–slip and measurement

Static friction exceeds kinetic: a fibre fringe or yarn that sticks, builds force, then slips in a judder — **stick–slip** — producing the saw-tooth force traces of fibre withdrawal tests, squeak in fabrics, and uneven drafting when rollers grip intermittently. Measurement methods: the **capstan method** itself (measure T_1, T_2, θ and invert the equation), inclined-plane and sled tests for fabrics, fibre-fringe withdrawal for fibre–fibre values, and single-fibre crossed-cylinder rigs in research. Typical working values: μ ≈ 0.2–0.35 for fibre on polished metal/ceramic with finish, higher fibre-to-fibre (cotton on cotton ≈ 0.3+), and wool scale-directed values differing by direction — next section.

4. Fibre friction at work

- **Cohesion and drafting:** spinning works because fibre–fibre friction lets roller drafting control a floating fibre mass; too little (over-lubricated) and fibres slip uncontrolled — slubs and thick places; too much and drafting force peaks snap fibres. Carding, combing and roving all budget friction.
- **Yarn and fabric integrity:** friction between fibres is yarn strength (short fibres rely on it entirely) and fabric resistance to slippage at seams; finishes that cut friction (softeners) predictably weaken

seam slippage resistance.

- **Wool's directional frictional effect (DFE):** scales make μ lower root-to-tip ("with" the scales) than tip-to-root. Under the repeated small movements of washing, each fibre migrates root-first and cannot return — the ratchet that is **felting**: irreversible entanglement shrinkage of wool. Anti-felt treatments (chlorination, resin coating) work by blunting the scales — friction engineering at the surface.
- **Processing tensions:** every guide in Topics on winding, warping and knitting multiplies or relaxes tension by capstan factors; machine settings assume a μ, and a finish change silently retunes the whole line.

5. Key points

- F = μN is a first approximation for fibres; F = aN^n with compliant contact, velocity, moisture and finish effects is the fuller truth.
- Capstan: $T_2 = T_1 e^{(\mu\theta)}$ — 180° at μ = 0.25 multiplies tension 2.19×; wrap angle is the strongest lever.
- Stick–slip separates static from kinetic friction and explains judder, squeak and erratic drafting.
- Fibre–fibre friction is productive (cohesion, drafting control, yarn strength) and destructive (felting via wool's DFE) in equal measure.
- Spin finishes exist to hold μ inside a narrow process window; changing finish retunes tensions everywhere.

6. Practice

Answer in your own words; brief model answers follow.

1. A yarn wraps 360° around a guide with μ = 0.2. Find the tension ratio. Input tension is 8 cN: find the output.
2. A technician halves the total wrap in a yarn path from 540° to 270° at the same μ = 0.28. By what factor does the tension multiplication change?
3. Why do Amontons' laws under-predict friction changes when normal load on a fibre bundle is doubled?
4. Explain, via directional friction, why a wool sweater felts in a hot agitated wash but a polyester one does not, and how "superwash" wool resists.
5. A mill changes spin finish and winding breaks rise though all tension settings are unchanged. Give the friction-based explanation and the first measurement you would make.

6. State the capstan condition under which a small brake force of 2 N can hold a load of 100 N (μ = 0.3): how much wrap is needed?

Answer key

1. θ = 2π = 6.283 rad; ratio = e^(0.2 × 6.283) = e^1.257 = 3.51. Output = 8 × 3.51 = 28.1 cN.
2. Old: e^(0.28 × 9.425) = e^2.639 = 14.0. New: e^(0.28 × 4.712) = e^1.319 = 3.74. The multiplication falls by a factor 14.0/3.74 ≈ 3.7× — halving the angle divides by the square root of the original factor (exponent halves).
3. Because real contact area is not fixed: doubling the load flattens soft fibres and enlarges true contact and interlocking, so friction grows less (or occasionally more) than proportionally — hence F = aN^n with n ≠ 1. Amontons assumes rigid, already-conforming surfaces.
4. Wool scales point root→tip, so μ is lower sliding with the scales than against them; agitation walks each fibre root-first into the mass and the reverse motion is locked — entanglement accumulates as felt. Polyester has a smooth, scale-free surface: no directional effect, no ratchet, no felting. Superwash chlorination erodes the scale edges (and resin coats them), collapsing the directional difference.
5. The new finish changed μ at every guide and roller; by the capstan law all path tensions rescaled while the settings stood still — higher μ means higher delivered tension and more breaks. First measurement: capstan-method μ of yarn against the actual guide material, old finish vs new.
6. Need e^(μθ) ≥ 50 → μθ ≥ ln 50 = 3.91 → θ ≥ 3.91/0.3 = 13.0 rad ≈ 2.1 turns. Two full turns plus a little — the bollard principle in numbers.
